Coreference Resolution

Relations between words

- Syntactic
- Semantic

in one sentence

Efforts in communication

- **Vangipurapu Venkata Sai Laxman** is an Indian former international cricketer and a former cricket commentator and pundit. **Vangipurapu Venkata Sai Laxman** played as a middle-order batsman in Test cricket. **Vangipurapu Venkata Sai Laxman** is widely regarded as one of the greatest batsmen in the history of test cricket. **Vangipurapu Venkata Sai Laxman** is currently the Head of Cricket at the National Cricket Academy (NCA). **Vangipurapu Venkata Sai Laxman** is the head coach of the India Under-19 and India A teams. **Vangipurapu Venkata Sai Laxman's** nickname is Very Very Special Laxman.

https://en.wikipedia.org/wiki/VVS_Laxman

Efforts in communication

- **Vangipurapu Venkata Sai (VVS) Laxman** is an Indian former international cricketer and a former cricket commentator and pundit. **VVS Laxman** played as a middle-order batsman in Test cricket. **VVS Laxman** is widely regarded as one of the greatest batsmen in the history of test cricket. **VVS Laxman** is currently the Head of Cricket at the National Cricket Academy (NCA). **VVS Laxman** is the head coach of the India Under-19 and India A teams. **VVS Laxman's** nickname is Very Very Special Laxman.

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- **Vangipurapu Venkata Sai (VVS) Laxman** is an Indian former international cricketer and a former cricket commentator and pundit. **He** played as a middle-order batsman in Test cricket **and** is widely regarded as one of the greatest batsmen in the history of test cricket. **He** is currently the Head of Cricket at the National Cricket Academy (NCA), **and** the head coach of the India Under-19 and India A teams. **His** nickname is Very Very Special.

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Terminology

- **Referent and Co-referents**

  Laxman is a former international cricketer. He played as a middle-order batsman in Test cricket and represented India in more than 100 Test matches. In 2011, Laxman was awarded the Padma Shri award, India's fourth highest civilian award. ....

  The 2001 Test at Eden Gardens, Kolkata against Australia was only the third time in the history of cricket that a team had managed to win a Test after being forced to follow on. It has become one of the most celebrated tales of Indian cricket, and the innings is ranked the sixth best Test innings ever by Wisden Cricketers' Almanack.

  Laxman's performance was of enormous consequence: India had been on the brink of an innings defeat but went on to win the Test and the series, denying Steve Waugh's conquest of the "final frontier".

  - Group-1: **Laxman, He, a middle-order batsman**
  - Group-2: **India, Indian**
  - Group-3: **[The 2001 Test at Eden Gardens, Kolkata against Australia], It, the Test**
  - Group-4: **Steve Waugh**

What about "India's" as in *India's fourth highest civilian award?*

NPs as Entities

- **Definite vs Indefinite**
  - Sapiens: A Brief History of Humankind is **a book** by Yuval Noah Harari. **It** was published in 2014 and **his** first bestseller. The graphic novel version of **the book** was published simultaneously in several languages.

Why Co-reference resolution is hard?

1. Ram wanted to meet Shyam urgently.
2. **They** were planning a trip to Kochi.
3. So **he** called **him**.
4. But **he** could not answer **the call**.
5. **He** was helping **his sister**, Gita, set up **her** business, **which** Ram did not know.
6. As Ram is **an impatient person**, **he** decided to go alone.
7. **He** blamed Shyam for spoiling **the plan**.
8. **He** liked Fort Kochi a lot.
9. **He** also took a ride in Kochi Water Metro.
10. **The metro** connects Kochi with some of **its** islands.
11. **It** is also connected with **the Kochi Metro**.
12. But **he** missed Shyam.
13. **He** blamed **himself** for **his** impatience.

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Domain and Co-references

- News articles
- Wikipedia articles (person vs event)
- Scientific documents